

Prompt Engineering and Large Language Models

Final Examination

Whole course, weighted towards Weeks 6–13 · 40 questions $\times$ 2.5 points = 100 points, 60% of the course grade

Name: _____ Student No: _____ Date: _____

Instructions. This paper contains 40 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 90 minutes. No notes or electronic devices.

1. Which best describes what a large language model does?
 - A) Retrieves the most relevant stored fact
 - B) Produces a probability for every possible next token, then one is selected
 - C) Applies rules written by its developers
 - D) Searches the internet and summarises the results
2. Why is hallucination not simply a bug awaiting a fix?
 - A) The models are still too small
 - B) There is no step in the operation at which truth is checked; plausibility is what is optimised
 - C) Training data has not yet been cleaned
 - D) Users write insufficiently precise prompts
3. In which situation is hallucination **most** likely?
 - A) A widely documented topic in English
 - B) A specific citation on a thinly covered topic
 - C) A request for a summary of provided text
 - D) A request to rewrite a paragraph more simply
4. A question is phrased “Explain why X causes Y”, when in fact X does not cause Y. What is the likely result?
 - A) The model corrects the premise
 - B) The model produces an explanation, accepting the premise
 - C) The model refuses to answer
 - D) The model reports low confidence
5. Why does the same content cost more tokens in Turkish than in English?
 - A) Turkish words are longer on average
 - B) Agglutinative stacking produces assembled words that appeared rarely in training
 - C) Turkish requires a separate tokenizer pass
 - D) Turkish uses more punctuation
6. An assistant is asked to count letters in a word and gets it wrong. Why?

- A) The arithmetic exceeded its capability
 - B) It never perceived individual letters — it operates on tokens
 - C) The context window overflowed
 - D) The temperature was set too high
7. Which is **not** held in the context window?
- A) The conversation so far
 - B) Attached documents
 - C) Hidden instructions added by the product
 - D) The model's training data
8. What does it mean that these systems are not deterministic?
- A) They cannot perform arithmetic
 - B) The same prompt may produce different answers on different runs
 - C) They cannot be tested
 - D) Their output depends on the time of day
9. Why is repeating a question and receiving a similar answer weak evidence of correctness?
- A) The model remembers its previous answer
 - B) Consistency measures how strongly a pattern is represented, not whether it is true
 - C) The second answer is always shorter
 - D) The context window has changed
10. Which best states the relationship between plausibility and truth in these systems?
- A) They are equivalent
 - B) They correlate, because training text was mostly written by people trying to be accurate
 - C) They are unrelated
 - D) Truth is checked and plausibility is a by-product
11. Adding “be accurate and use only reliable sources” to a prompt about a thinly covered topic will most likely:
- A) Improve factual accuracy
 - B) Produce a refusal
 - C) Make the answer sound more authoritative without improving it
 - D) Trigger a web search
12. Which of the four levers should be used when the output has the right content in the wrong shape?
- A) Context
 - B) Task
 - C) Format
 - D) Examples
13. Which is a well-formed negative constraint?
- A) Do not be vague
 - B) Do not write anything unprofessional

- C) Do not cite any case or regulation by name
 - D) Do not disappoint the reader
14. An answer about Turkish procedure describes, accurately, the procedure of another country. This is:
- A) Fabrication
 - B) Displacement
 - C) Omission
 - D) Silent failure
15. A legal memo contains a case name in correct citation format for a case that was never decided. This is:
- A) Fabrication
 - B) Displacement
 - C) Over-reading
 - D) Indirect injection
16. Which failure is hardest to detect outside your own area of expertise, and why?
- A) Fabrication — because citations look real
 - B) Displacement — because foreign systems are unfamiliar
 - C) Omission — because there is no incorrect statement to identify
 - D) Overconfidence — because tone is uniform
17. A grounded assistant answers a question your documents do not cover, producing a confident response with no quotation. This is:
- A) Retrieved the wrong passage
 - B) Over-read the passage
 - C) Answered from memory
 - D) Silent failure
18. Your source says “the committee *may* require additional evidence”. The assistant reports “the committee *requires* additional evidence”. This is:
- A) Fabrication
 - B) Over-reading the passage
 - C) Answered from memory
 - D) Displacement
19. A grounded answer quotes a real sentence from your document, but the sentence comes from an unrelated section and does not support the claim. This is:
- A) Fabrication
 - B) Retrieved the wrong passage
 - C) Omission
 - D) Leakage
20. A three-step workflow receives faulty input at step one and produces a complete, confident, well-formatted output. Nothing reports an error. This is:
- A) Wrong stopping point

- B) Silent failure
 - C) Context overflow
 - D) Direct injection
21. What do all nine failure modes covered in this course have in common?
- A) They occur only in long conversations
 - B) They are invisible in the output
 - C) They can be prevented by better prompting
 - D) They occur only in grounded systems
22. A generated browser tool produces a subtly wrong number, consistently, behind a clean interface. Why is consistency dangerous here?
- A) It makes the error harder to reproduce
 - B) It makes the tool appear reliable
 - C) It increases the token cost
 - D) It causes the tool to fail eventually
23. Which check would most reliably expose a poisoned document summary?
- A) Asking the model whether it was manipulated
 - B) Regenerating the summary several times
 - C) Checking each claim against a quoted passage in the source
 - D) Reducing the temperature
24. What is the most significant thing lost when a chatbot is replaced by an agent?
- A) Response speed
 - B) The unconscious checking you performed on every intermediate output
 - C) The ability to attach documents
 - D) Access to the context window
25. If each step is 95% reliable, approximately what is end-to-end reliability over ten steps?
- A) 95%
 - B) 80%
 - C) 60%
 - D) 40%
26. Why is compounding error worse than independent random error?
- A) Errors are reported only at the end
 - B) Each step consumes the previous step's output, so early errors are inherited
 - C) Agents use more tokens per step
 - D) Later steps are inherently less reliable
27. Which control does **not** rely on the system behaving as intended?
- A) Instructing it to be careful
 - B) Asking it to verify its own output
 - C) Connecting only the narrowest tools required

- D) Requesting quotations for each claim
28. Where should a human be placed in an AI workflow?
- A) At the end, reviewing the output
 - B) Before the irreversible action
 - C) At the start, writing the prompt
 - D) Wherever capacity allows
29. Which is **not** a criterion for a good first automation candidate?
- A) The task is genuinely repetitive
 - B) The answers exist in documents you can point at
 - C) A wrong answer is recoverable
 - D) The task is the most intellectually difficult in your field
30. Why should the harmed person be named specifically rather than as a category?
- A) Institutional policy requires it
 - B) Specificity changes the design; abstraction does not
 - C) It shortens the risk assessment
 - D) Categories cannot be defined precisely
31. In a well-designed AI workflow, most decisions concern:
- A) Which model to use
 - B) What the system must not do
 - C) How to phrase the prompt
 - D) Which tool provider to select
32. A Domain Pack improves all of the following **except**:
- A) Local vocabulary
 - B) Recognisable structure
 - C) Conventional restraint
 - D) Factual accuracy
33. Why can a Domain Pack increase risk as well as usefulness?
- A) It consumes context window space
 - B) Professional formatting makes wrong content more persuasive
 - C) It slows the system down
 - D) It prevents the system from refusing
34. Why do professional structures such as SOAP and IRAC help when working with these systems?
- A) They are shorter than free prose
 - B) Each makes a particular kind of error visible
 - C) They are familiar to the model
 - D) They reduce token usage
35. What is the root cause of prompt injection?

- A) Insufficient safety fine-tuning
 - B) The model receives one stream of tokens and cannot reliably separate instructions from data
 - C) Context windows are too small
 - D) Users upload untrusted files
- 36.** Why is indirect injection more serious for an ordinary user than direct injection?
- A) It is technically harder to perform
 - B) The malicious instruction is hidden in content the user never sees, so there is no point at which they could refuse
 - C) It cannot be defended against at all
 - D) It only affects grounded systems
- 37.** Which attacker goal requires an agent rather than a chatbot?
- A) Changing the content of a summary
 - B) Extracting other material from the context
 - C) Causing an action such as sending or deleting
 - D) Revealing the system instructions
- 38.** Which statement about anonymisation is correct?
- A) Removing names is sufficient
 - B) A rare condition, a date and a small department can identify someone even with names removed
 - C) Anonymised data may be uploaded to any service
 - D) Anonymisation is required only for research
- 39.** Under this course's account of academic integrity, the central question is:
- A) Whether AI was used at all
 - B) What the assessment is measuring, and what you are certifying by submitting
 - C) Whether the use can be detected
 - D) How much of the text was generated
- 40.** What is the strongest argument for occasionally drafting work yourself?
- A) It is faster for short documents
 - B) The judgment that lets you evaluate generated output is built by doing the work
 - C) Institutional rules require it
 - D) Generated drafts are usually poor